Tanqiu Jiang

tanjiang@cs.stonybrook.edu | Cell: (610)-674-8718 | tanqiujiang.github.io

Research Interests

AI Safety and Trustworthy Machine Learning.
Security and robustness of LLMs and multimodal agents; adversarial evaluation and defenses against jailbreaks, prompt injection, and long-horizon attacks; privacy-preserving generative modeling.
Broader interests include safe agent learning and self-improvement and reliable AI for automated research.

Education

Stony Brook University — Stony Brook, NY
Ph.D. in Computer Science *Advisor: Dr. Ting Wang* — *Aug 2023 – Present*

University of Rochester — Rochester, NY
Master of Science in Data Science — *Aug 2020 – Dec 2021*

Lehigh University — Bethlehem, PA
Bachelor of Science in Computer Engineering (Minor: Data Science) — *Aug 2016 – May 2020*

Publications

Peer-Reviewed Conference Papers

[C1] **Jiang, T.**, Wang, Y., Liang, J., & Wang, T. (2026). AgentLAB: Benchmarking LLM Agents against Long-Horizon Attacks. In *International Conference on Machine Learning* (**ICML 2026**). [pdf]
[C2] **Jiang, T.**, Liang, J., Zhu, R., Zhou, J., Ma, F., & Wang, T. (2026). Dynamic Token Reweighting for Robust Vision-Language Models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition* (**CVPR 2026**). [pdf]
[C3] Wang, Y., **Jiang, T.**, Liang, J., Fleming, C., & Wang, T. (2026). MAGE: Safeguarding LLM Agents against Long-Horizon Threats via Shadow Memory. In *ACM Conference on Computer and Communications Security* (**CCS 2026**). [pdf]
[C4] Liang, J., **Jiang, T.**, Wang, Y., Zhu, R., Ma, F., & Wang, T. (2026). AutoRAN: Weak-to-Strong Jailbreaking of Large Reasoning Models. In *Annual Meeting of the Association for Computational Linguistics* (**ACL 2026**). [pdf]
[C5] Liang, J., Wang, Y., Li, C., Zhu, R., **Jiang, T.**, Gong, N., & Wang, T. (2026). GraphRAG Under Fire. In *IEEE Symposium on Security and Privacy* (**S&P 2026**). [pdf]
[C6] Liang, J., Wang, Y., **Jiang, T.**, & Wang, T. (2026). RASA: Routing-Aware Safety Alignment for Mixture-of-Experts Models. In *Conference on Empirical Methods in Natural Language Processing* (**EMNLP 2026**). [pdf]
[C7] Liang, J., Wang, Y., **Jiang, T.**, & Wang, T. (2026). LaCache: Robust Semantic Caching for LLM Serving. In *Conference on Neural Information Processing Systems* (**NeurIPS 2026**).
[C8] **Jiang, T.**, Wang, Z., Liang, J., Li, C., Wang, Y., & Wang, T. (2025). RobustKV: Defending Large Language Models against Jailbreak Attacks via KV Eviction. In *International Conference on Learning Representations* (**ICLR 2025**). [pdf]
[C9] **Jiang, T.**, Li, C., Ma, F., & Wang, T. (2025). RAPID: Retrieval Augmented Training of Differentially Private Diffusion Models. In *International Conference on Learning Representations* (**ICLR 2025**). [pdf]
[C10] **Jiang, T.**, Bendre, S. K., Lyu, H., & Luo, J. (2021). From Static to Dynamic Prediction: Wildfire Risk Assessment Based on Multiple Environmental Factors. In *2021 IEEE International Conference on Big Data* (**IEEE Big Data**). [pdf]
[C11] **Jiang, T.** & Xiong, Z. (2021). Rule-Based Approach to the Automatic Detection of Individual Tree Crowns in RGB Satellite Images. In *2021 IEEE International Conference on Computer Science, Artificial Intelligence and Electronic Engineering (IEEE-CSAIEE)*, pp. 132–135. [pdf]
[C12] Wang, X., **Jiang, T.**, & Cai, H. II. (2021). Human Epithelial-2 Cell Image Classification Using Deep Unsupervised Learning and Gradient Boosting Trees. In *Proc. SPIE 11601, Medical Imaging 2021*. [pdf]
[C13] **Jiang, T.**, Li, Y., Lin, H., Ruan, Y., & Woodruff, D. P. (2020). Learning-Augmented Data Stream Algorithms. In *8th International Conference on Learning Representations* (**ICLR 2020**), Addis Ababa, Ethiopia. [pdf]

Journal Papers

[J1] Tian, X., Liu, Y., Zhang, K. L., Litt, S., Shin, K., & **Jiang, T.** (2026). Do Artificial Intelligence Clients Speak Like Human Clients? Exploring GPT-4's Content-Level Performance in Counseling Role-Play. *Journal of Counseling & Development*, 104(1), 14–26.
[J2] Tian, X., Liu, Y., Litt, S., Shin, K., & **Jiang, T.** (2025). Artificial Clients, Genuine Interactions? Assessing ChatGPT-4's Process-Level Performance in Clinical Simulations Using Qualitative Content Analysis. *Counselling and Psychotherapy Research*, 25(4), e70063.

Manuscripts Under Review & Preprints

[P1] **Jiang, T.**, Bai, M., Shahriar, S., Du, W., & Swamy, S. (2026). PerMIT: Securing Persistent Memory for LLM Agents via Machine-Checked Integrity Theorems and Scalable Community Trust. Manuscript under review at the *International Conference on Learning Representations* (**ICLR 2027**). [pdf]
[P2] Zhu, R., **Jiang, T.**, Ma, F., & Wang, T. (2026). Understanding and Mitigating the Fragility of Self-Rewarding Optimization. Manuscript under review at the *International Conference on Learning Representations* (**ICLR 2027**).
[P3] **Jiang, T.**, Bai, M., Pappas, N., Qi, Y., & Swamy, S. (2026). Cross-Modal Content Optimization for Steering Web Agent Preferences. Manuscript under review at the *20th Conference of the European Chapter of the Association for Computational Linguistics* (**EACL 2027**). [pdf]
[P4] Zhu, R., Wang, Y., **Jiang, T.**, Liang, J., & Wang, T. Self-Improving Model Steering. *arXiv preprint*. [pdf]

Experience

Amazon AWS Seattle, WA
Applied Scientist Intern, AWS AI Fundamental Research *May 2026 – Aug 2026*

- Led the design and development of PerMIT [P1], a persistent-memory framework that protects LLM agents against injection and poisoning through information flow controls (IFC) while leaving agent reasoning, planning, and tool use unrestricted.
- Formalized the safety guarantee of PerMIT in Lean 4 and machine-checked the complete memory-operation interface
- Extended the guarantees to shared memory through accumulated endorsements, supplying a trust layer for existing frameworks such as Open Knowledge Format (OKF) and proving that safety and utility co-scale as communities grow.
- Achieved 0% attack success across all evaluated backbones and LLM verifiers on Memory attack benchmarks

Applied Scientist Intern, Agentic AI Lab *May 2025 – Sep 2025*

- Developed cross-modal adversarial attack framework to identify vulnerabilities in VLM-based web agents, achieving up to 71% preference manipulation rate across GPT-4.1, Qwen-2.5VL, and Pixtral-Large models.
- Implemented PGD-based visual perturbation attacks exploiting CLIP encoder vulnerabilities, achieving $>$50% black-box transfer success while maintaining imperceptibility (ϵ=8/255).
- Designed iterative textual refinement algorithms leveraging RLHF-induced biases to steer agent preferences through semantically plausible content modifications.
- Conducted comprehensive evaluation on ControlNet-100k and VisualWebArena benchmarks, demonstrating 3.5$\times$ improvement over baseline methods while evading detection; findings released as an arXiv preprint [P3].

Stony Brook University Stony Brook, NY
Research Assistant, Advisor: Dr. Ting Wang *Aug 2023 – Present*

- Conduct research in AI safety, focusing on the safety and robustness of large language models and LLM-based agents.
- Lead multiple first-author research projects spanning LLM jailbreak defense (RobustKV, ICLR 2025), differentially private generative modeling (RAPID, ICLR 2025), adversarial robustness of vision-language models (DTR, CVPR 2026), and long-horizon attack benchmarking for LLM agents (AgentLAB, ICML 2026).
- Collaborate on research addressing memory-based defenses for LLM agents (MAGE, CCS 2026), jailbreaking of large reasoning models (AutoRAN, ACL 2026), adversarial attacks on GraphRAG systems (S&P 2026), and routing-aware safety alignment for mixture-of-experts models (RASA, EMNLP 2026).

Pennsylvania State University State College, PA
Research Assistant *Aug 2022 – May 2023*

- Led a research project developing "Unlearnable Examples" aimed at minimizing the performance of contrastive learning models by introducing adversarial noise into training images.
- Attended and presented in weekly individual and group meetings with Dr. Ting Wang, Dr. Dongwon Lee, and lab mates.

University of Rochester, Goergen Institute for Data Science Rochester, NY
Teaching Assistant *Jan 2021 – Dec 2021*

- Held office hours averaging over 2 hours weekly to assist students with Python, Linux, SQL, PySpark, and R.
- Collaborated with Professors Lloyd Palum and Brendan Mort to test and refine course materials on Databricks.
- Conducted review sessions to reinforce course material and prepare students for assessments.
- Graded assignments and projects, providing constructive feedback to students.

Honors and Awards

Graham Endowed Fellowship ($4,000), Pennsylvania State University, Aug 2022
NSF Student Travel Grant ($500), IEEE Big Data Conference, Dec 2021
Silver Medal (Top 3%, 16th/541), Kaggle: Google Landmark Retrieval Challenge, 2020